

Climate Change & Its Global Impact
Understanding Our Planet's Crisis and Path Forward
Environmental Science Team - November 2024

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Agenda

Introduction to Climate Change

Primary Causes and Drivers

Ecosystem Disruption and Biodiversity Loss

Rising Sea Levels and Coastal Threats

Economic Consequences and Human Impact

Renewable Energy Solutions

Global Initiatives and Agreements

Case Studies and Success Stories

Future Projections and Scenarios

Key Takeaways and Call to Action

Introduction to Climate Change

Climate change refers to long-term shifts in global temperatures and weather patterns

Primarily driven by human activities since the Industrial Revolution

Global average temperature has risen by approximately 1.1°C since pre-industrial

Scientific consensus attributes over 95% probability to human influence

Manifestation includes extreme weather events, melting ice caps, and ecosystem disruption

Primary Causes and Drivers

Fossil Fuel Combustion (75% of emissions)

Deforestation and Land Use Changes

Industrial Processes and Agriculture

Methane Emissions from Livestock

Cement Production and Manufacturing

Carbon dioxide (CO₂) - 76% of greenhouse gases

Methane (CH₄) - 16% of greenhouse gases

Nitrous oxide (N₂O) - 6% of greenhouse gases

Fluorinated gases - 1% of greenhouse gases

Black carbon and particulate matter

Ecosystem Disruption and Biodiversity Loss

Coral reefs experiencing 50% mortality due to ocean warming and acidification

Arctic sea ice decline of 13% per decade since 1980s

Species extinction rates 100-1,000 times above natural background

Amazon rainforest showing signs of tipping point with increased droughts

Migration pattern disruptions affecting predator-prey relationships

Rising Sea Levels and Coastal Threats

Global sea level rose 20cm since 1900, accelerating to 3.3mm/year currently

Thermal expansion accounts for 40% of rise, ice sheet melt 60%

Coastal flooding affects 680 million people globally

Small island nations face existential threats - Maldives, Tuvalu at risk

Economic losses from storm surge and flooding exceed \$1 trillion annually

Economic Consequences and Human Impact

Climate disasters cost global economy \$268 billion in 2022

Agricultural yields projected to decline 17% by 2050 in tropical regions

Health impacts from heat stress and air pollution affect 10 million annually

Climate migration displacing 21.5 million people yearly from weather events

GDP losses estimated at 10% globally if warming exceeds 4°C

Renewable Energy Solutions

Solar Power - 28% of global electricity generation growth

Wind Energy - 70% cost reduction since 2010

Battery Storage - Grid-scale solutions emerging

Hydroelectric - Largest renewable source currently

Geothermal and Tidal - Emerging technologies

Solar capacity increased 20-fold since 2010

Wind power prevents 1.1 billion tons CO₂ emissions annually

Battery costs dropped 89% from 2010 to 2022

Renewables provide 30% of global electricity

Potential for 90% clean electricity by 2050

Global Initiatives and Agreements

Future Projections and Scenarios

Current policies trajectory: 2.7°C warming by 2100

1.5°C pathway requires 43% emissions reduction by 2030

Sea level rise: 26-86cm by 2100 depending on mitigation

Extreme heat days could increase 5x in some regions

Positive outlook: Renewables and technology offer hope for stabilization

Key Takeaways

Climate crisis is the defining challenge of our generation

Immediate action required: emissions down 45% by 2030

Renewable energy transition is economically viable and accelerating

Global cooperation through treaties shows measurable progress

Individual and collective action can still prevent worst-case scenarios

Together for a Sustainable Future
Questions & Discussion